

Shangjie (Henry) Zheng

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TECHNICAL SKILLS

- **Programming Languages:** Java, Python, C++, C, SQL, MATLAB, JavaScript, TypeScript, HTML, CSS
- **Databases & Data Streaming:** MySQL, PostgreSQL, SQLite, MongoDB, Amazon S3, Redis, Kafka, BullMQ, Hibernate
- **Frameworks & Platforms:** Spring Boot, Django, React, Node.JS, Express, AWS, Linux, Zephyr RTOS, CAN, J1939, Modbus
- **Tools & Libraries:** Docker, Kubernetes, Terraform, Git/GitHub, Postman, Maven, Apache JMeter, Nginx, spaCy, NLTK

PROFESSIONAL EXPERIENCE

Lion Energy LLC. American Fork, UT
Embedded Systems Engineer, Controls Software | C, Zephyr RTOS, MATLAB, Simulink, J1939 Jan. 2026 – Present

- Architected and implemented a modular embedded application layer in C on Zephyr RTOS for Energy Storage Systems (ESS), enabling seamless system-level control of up to 6 power distribution units through deterministic task scheduling and structured fault management.
- Developed and optimized a CAN-based messaging subsystem (J1939) serving as a unified communication service layer across 7+ ESS functional domains, enabling distributed peer-to-peer and broadcast communication, multipacket transport, and OTA update capabilities.
- Implemented cloud-connected embedded software features within the Cloud Controller, extending application modules and validating end-to-end communications across AWS IoT (MQTT), proprietary board-to-board communications, and CAN-based embedded devices.

Southern Methodist University Dallas, TX (Remote)
Research Collaborator | Python, Data Pipelines, NLTK, spaCy Aug. 2023 – Present

- Developed Python pipelines to collect, validate, and organize tax-domain documents from online sources into structured research datasets.
- Built an interactive annotation interface to present domain-specific text, capture classification decisions, and automate label collection.
- Evaluated researcher-provided classification models by comparing performance and analyzing results to support research decisions.

Lion Energy LLC. American Fork, UT
Control Systems Engineer | C, MATLAB, Simulink, J1939, Zephyr RTOS Dec. 2024 – Jan. 2026

- Designed and optimized software components for Energy Storage System, focusing on reliability and performance improvements.
- Led the design and implementation of a DSP module within the real-time telemetry pipeline, improving aggregate data accuracy by 35%.
- Built a MATLAB-based test automation framework integrated with Simulink Test API, reducing manual test creation time by 90%.

Lion Energy LLC. American Fork, UT
Control Systems Engineer Intern | C, MATLAB, Simulink, MCU, CAN Jul. 2024 – Dec. 2024

- Contributed to the development and evaluation of embedded control system software for Battery Management System (BMS).
- Refined SoC estimation through end-of-charge/discharge state Amp-Volt mapping optimization, increasing accuracy by over 30%.
- Redesigned cell balancing algorithm to discharge optimal balancing current from target cells, speeding process from days to hours.

SELECTED PROJECTS

Riivr App, Distributed Backend Social Platform American Fork, UT
Node.js, TypeScript, MongoDB, Redis, BullMQ, AWS (EC2, S3, ElastiCache, CodeDeploy), Terraform Spring, 2026

- Architected a domain-driven backend spanning 9 bounded contexts, 60+ REST endpoints, and 10 persistence models, integrating JWT authentication in a layered pipeline (routes, middleware, controllers, services), enforcing modular boundaries and validations.
- Developed a cache-first Redis-over-MongoDB strategy, reducing latency and database pressure with write-through consistency.
- Leveraged BullMQ-backed asynchronous job queues to offload non-blocking workflows and decouple background processing.
- Provisioned AWS infrastructure via Terraform across 15+ resources (multi-AZ, ALB/HTTPS, EC2, NAT, Bastion, ElastiCache).

NEXUS-OS, 32-bit x86 Operating System Lehi, UT
C, x86 Assembly, NASM, GCC, QEMU, FAT16 Summer, 2026

- Built a 32-bit x86 kernel with protected-mode bootstrap, GDT/IDT interrupts, virtual memory, timer-driven multitasking, ELF loading, and FAT16 filesystem, enabling isolated kernel/user execution, hardware exception handling and independent process address spaces.
- Implemented Ring 0/3 privilege separation, context switching, int 0x80 syscalls, providing controlled user-space access to kernel services.
- Integrated PS/2 keyboard input and VGA output into an interactive shell for launching user programs from a custom FAT16 filesystem.

EDUCATION

Southern Methodist University (SMU) May 2024
Master of Science in Computer Science GPA: 3.97/4.00

University of Missouri-Kansas City (UMKC) May 2021
Bachelor of Science in Mechanical Engineering GPA: 3.52/4.00